



Govt Graduate College for Women Baghbanpura , Lahore

BS Psychology 4-Year Degree Program

Biological basis of Behaviour

Assignment Topic: Neurochemistry

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Submitted by: Sidra Imtiaz

Submitted to: Miss Adeela

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Neurochemistry

Neurochemistry is the branch of science that deals with the chemical processes and substances that occur within the nervous system. It explores how neurotransmitters, hormones, and other molecules affect the function of neurons and how they contribute to behaviors, emotions, and cognitive functions.

Some key aspects of neurochemistry include:

1.Neurotransmitters: Chemical messengers that transmit signals between neurons, such as dopamine, serotonin, and acetylcholine.

2.Neuroplasticity: The brain's ability to adapt and change in response to experience, learning, and injury.

3.Synapses: The gaps between neurons where chemical signals are transmitted.

4.Hormones: Chemical messengers that regulate various bodily functions, such as stress response (cortisol) and mood regulation (thyroid hormones).

5.Brain regions: Different areas of the brain, such as the hippocampus (memory) and amygdala (emotions), have distinct neurochemical profiles.

6.Neurochemical imbalance: Alterations in neurochemical levels or function can contribute to various neurological and psychiatric conditions, such as depression, anxiety, and Parkinson's disease.

7.Neuropharmacology: The study of how drugs interact with neurochemical systems to produce therapeutic or adverse effects.

- **PREVIOUS CONCEPT:** Messages are sent electrically through neurons.
- **NEW CONCEPT:** B.Otto Lowei and Henry Dale proved messages were sent chemically.

Neuromodulators: Neuromodulators are chemicals that modulate neural activity and have a broader influence on neural circuits than traditional neurotransmitters. They can:

1. Modify synaptic strength.
2. Change neural excitability
3. Influence neural oscillations
4. Regulate gene expression
5. Modulate neuroplasticity

Examples of neuromodulators include:

1. Gases (e.g., nitric oxide, carbon monoxide)
2. Neurotrophic factors (e.g., BDNF, NGF)
3. Hormones (e.g., thyroid hormone, estrogen)

Neuromodulators play a crucial role in various physiological and psychological processes, including:

1. Mood regulation
2. Motivation and reward
3. Cognitive function

Characteristics of Neuromodulators:

1. Modulatory role: They don't directly transmit signals but instead modify the strength or nature of synaptic transmission.
2. Broad influence: Neuromodulators can affect multiple neurons, neural circuits, or brain regions.
3. Long-lasting effects: Their effects can persist for minutes, hours, or even days.
4. Diverse chemical structures: Neuromodulators include various chemicals, such as neurotransmitters, hormones, and gases (e.g., nitric oxide).
5. Multiple receptors: They often bind to multiple receptor subtypes, each with distinct effects.
6. Context-dependent actions: Their effects depend on the specific neural context, including the type of neurons, synaptic activity, and behavioral state.
7. Involvement in various functions: Neuromodulators participate in regulating mood, motivation, attention, memory, and other physiological processes.

8.Imbalance implications: Dysregulation of neuromodulators has been implicated in various neurological and psychiatric disorders.

9.Target for therapeutics: Neuromodulators are a focus for drug development, particularly for mood disorders and cognitive enhancement.

10.Complex interactions: Neuromodulators interact with other neuromodulators and neurotransmitters, creating complex neural circuits.

Neurohormone:Neurohormones are chemicals produced by neurons that have a hormonal effect on the body. They are synthesized in the brain and released into the bloodstream, where they can bind to receptors on target cells to regulate various physiological processes.

Examples of neurohormones:

1.Oxytocin: involved in social bonding, childbirth, and lactation

2.Vasopressin: regulates water balance and blood pressure

3.Melatonin: regulates sleep-wake cycles

4.Adrenocorticotrophic hormone (ACTH): stimulates the adrenal glands to produce cortisol

5. Thyroid-stimulating hormone (TSH): stimulates the thyroid gland to produce thyroid hormones

6.Prolactin: involved in lactation and sexual satisfaction

7.Neurosteroids: involved in mood regulation and neuroprotection

8.Growth hormone: regulates growth and development

9.Follicle-stimulating hormone (FSH): regulates reproductive processes

10.Luteinizing hormone (LH):regulates reproductive processes

Neurohormones play a crucial **role** in various physiological processes, including:

- Regulation of stress response

- Mood regulation
- Sleep-wake cycles
- Reproductive processes
- Metabolism and energy balance
- Growth and development
- Lactation and childbirth

Characteristics of Neurohormone:

- 1. Produced by neurons:** Neurohormones are synthesized and released by neurons in the brain.
- 2. Hormone-like actions:** They have similar effects to hormones produced by endocrine glands, but with a more localized action.
- 3. Regulate various functions:** Neurohormones influence mood, motivation, appetite, sleep, stress response, and other physiological processes.
- 4. Bind to specific receptors:** They interact with specific receptors on target cells, triggering intracellular signaling cascades.
- 5. Short- and long-term effects:** Neurohormones can have both rapid and sustained effects, depending on the context.
- 6. Regulated by feedback loops:** Their production and release are often controlled by negative feedback mechanisms to maintain homeostasis.
- 7. Interact with other neurochemicals:** Neurohormones can modulate the activity of other neurotransmitters and neuromodulators.
- 8. Involved in neuroplasticity:** They contribute to changes in neural circuits and synaptic strength.
- 9. Dysregulation implications:** Imbalances or abnormalities in neurohormone systems have been linked to various neurological and psychiatric disorders.
- 10. Target for therapeutic interventions:** Neurohormones are a focus for drug development and therapeutic strategies.

